

Conversation: Applied linear algebra final exam preparation
 Exported from Claude.ai
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--- You ---

I need help for prepparing for my final exam for applied linear algebra where the goal is to demonstrate understanding of different topics

--- Claude ---

I'd love to help you prep! To make this as useful as possible, a few quick questions:

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--- You ---

Q: What format is the exam?

A: Free response / proofs

Q: Which topics do you want to focus on? (Select all that apply)

A: Matrix operations & systems of equations, Eigenvalues & eigenvectors, Vector spaces, bases & dimension, Linear transformations

Q: How do you want to study?

A: Concept explanations + examples

--- Claude ---

Great choices. Those four topics form the core of most applied linear algebra courses, and they're deeply interconnected. Let me walk through each one clearly with examples.

1. Matrix Operations & Systems of Equations

Core Ideas

A system of linear equations $Ax = b$ can be solved by row reducing the augmented matrix $[A \mid b]$ to row echelon form (REF) or reduced row echelon form (RREF).

Three possible outcomes:

- **Unique solution** -- one pivot per variable, consistent system
- **Infinitely many solutions** -- free variables exist (underdetermined)
- **No solution** -- inconsistent (a row like $[0 \ 0 \mid 5]$ appears)

Key Operations

- **Row reduction (Gaussian elimination):** Systematically eliminate entries below each pivot
- **Matrix multiplication:** $(AB)_{ij} = \text{dot product of row } i \text{ of } A \text{ with column } j \text{ of } B$
- **Inverse:** A^{-1} exists iff $\det(A) \neq 0$ iff A has full rank

Example

Solve the system:

```

$$x + 2y - z = 4$$

$$2x + y + z = 6$$

$$x - y + 2z = 1$$

```

Augmented matrix and row reduce:

```

$$\begin{array}{ccc|c} [1 & 2 & -1 & 4] \\ [2 & 1 & 1 & 6] \\ [1 & -1 & 2 & 1] \end{array} \rightarrow \begin{array}{ccc|c} [1 & 2 & -1 & 4] \\ [0 & -3 & 3 & -2] \\ [0 & -3 & 3 & -3] \end{array} \begin{array}{l} \\ (R2 - 2R1) \\ (R3 - R1) \end{array}$$

---

R3 - R2 gives  $[0 \ 0 \ 0 \mid -1]$ , which is a contradiction -- **no solution**.

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## ## 2. Eigenvalues & Eigenvectors

### ### Core Ideas

A scalar  $\lambda$  is an **eigenvalue** of A if there exists a nonzero vector v such that:

$$Av = \lambda v$$

That vector v is the corresponding **eigenvector**. Geometrically, A only stretches or flips v -- it doesn't rotate it.

### ### How to Find Them

1. Solve  $\det(A - \lambda I) = 0$  -- this is the **characteristic equation**
2. For each  $\lambda$ , solve  $(A - \lambda I)v = 0$  to find the eigenvectors

### ### Example

Let  $A = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$

**Step 1 -- Characteristic equation:**

$$\det(A - \lambda I) = (3 - \lambda)(2 - \lambda) - 0 = 0$$

$$\lambda^2 - 5\lambda + 6 = 0 \implies \lambda = 3 \text{ or } \lambda = 2$$

**Step 2 -- Eigenvectors:**

$$\text{For } \lambda = 3: (A - 3I)v = \begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix} v = 0 \implies v = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\text{For } \lambda = 2: (A - 2I)v = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} v = 0 \implies v = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

### ### Key Facts for Proofs

- The **trace** of A = sum of eigenvalues
- The **determinant** of A = product of eigenvalues
- Eigenvectors from **distinct eigenvalues** are always linearly independent
- A is diagonalizable iff it has n linearly independent eigenvectors

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## ## 3. Vector Spaces, Bases & Dimension

### ### Core Ideas

A **vector space** is a set V with addition and scalar multiplication satisfying 8 axioms (closure, associativity, identity, inverses, distributivity, etc.).

Important subspaces of a matrix A:

- **Column space**  $\text{Col}(A)$ : span of the columns of A
- **Null space**  $\text{Nul}(A)$ : all x such that  $Ax = 0$
- **Row space**: span of the rows of A

### ### Basis & Dimension

A **basis** is a set of vectors that is:

1. **Linearly independent**
2. **Spans** the space

The **dimension** is the number of vectors in any basis. This leads to the critical:

**Rank-Nullity Theorem:**  $\text{rank}(A) + \text{nullity}(A) = n$  (number of columns)

Where  $\text{rank} = \dim(\text{Col}(A))$  and  $\text{nullity} = \dim(\text{Nul}(A))$ .

### Example

For  $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$ :

Row reduce to find pivots in columns 1 and 2 -- rank = 2.

- **Col(A) basis:** columns 1 and 2 of the *original*  $A$ :  $\{[1,0,0], [2,1,0]\}$
- **Nul(A):** set  $x_3 = t$  (free), back-substitute:  $x_2 = -2t$ ,  $x_1 = -t$
- Nul(A) =  $\text{span}\{[-1, -2, 1]\}$
- **Check:** rank(2) + nullity(1) = 3 columns

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## ## 4. Linear Transformations

### Core Ideas

A function  $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$  is a **linear transformation** if:

1.  $T(u + v) = T(u) + T(v)$
2.  $T(cu) = cT(u)$

Every linear transformation can be represented as  $T(x) = Ax$  for some matrix  $A$ , called the **standard matrix** of  $T$ .

### Standard Matrix

The columns of  $A$  are just where  $T$  sends each standard basis vector:

$A = [T(e_1) \mid T(e_2) \mid \dots \mid T(e_n)]$

### Key Properties

- **One-to-one (injective):** Nul(A) =  $\{0\}$ , i.e. each  $x$  maps to a unique output
- **Onto (surjective):** Col(A) =  $\mathbb{R}^m$ , i.e. every  $b$  has a preimage
- For square  $A$ : one-to-one  $\iff$  onto  $\iff A$  invertible

### Example

$T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$  rotates vectors by  $90^\circ$  counterclockwise.

$T(e_1) = T([1,0]) = [0,1]$

$T(e_2) = T([0,1]) = [-1,0]$

Standard matrix:  $A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

To verify linearity:  $T(u+v) = A(u+v) = Au + Av = T(u) + T(v)$

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## ## How These Topics Connect

A useful mental map:

- **A system  $Ax = b$**  is consistent iff  $b$  is in Col(A) (Linear Transformations + Vector Spaces)
- **Eigenvalues** tell you whether  $A$  is invertible ( $\lambda = 0$  is an eigenvalue  $\iff$  singular matrix  $\iff$  nontrivial null space)
- **Diagonalization**  $A = PDP^{-1}$  turns repeated matrix powers into trivial scalar powers -- connecting all four topics

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Want me to drill deeper into any of these, work through proof strategies (like how to prove something is a subspace or show a transformation is linear), or move into practice problems?

--- You ---

lets narrow down on the idea of demonstrate understanding of the concepts of vector space and subspace

--- Claude ---

Let me break this down with exactly the kind of depth you'd need for a free response exam.

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## ## Vector Spaces

### ### The Definition (What You Must Know Cold)

A **vector space**  $V$  is a set with two operations -- addition and scalar multiplication -- satisfying these **10 axioms** for all  $u, v, w$  in  $V$  and scalars  $c, d$ :

| #  | Axiom                                                     |
|----|-----------------------------------------------------------|
| 1  | $u + v$ is in $V$ (closure under addition)                |
| 2  | $u + v = v + u$ (commutativity)                           |
| 3  | $(u + v) + w = u + (v + w)$ (associativity)               |
| 4  | There exists a zero vector $0$ such that $u + 0 = u$      |
| 5  | For each $u$ , there exists $-u$ such that $u + (-u) = 0$ |
| 6  | $cu$ is in $V$ (closure under scalar multiplication)      |
| 7  | $c(u + v) = cu + cv$                                      |
| 8  | $(c + d)u = cu + du$                                      |
| 9  | $c(du) = (cd)u$                                           |
| 10 | $1u = u$                                                  |

On an exam, you rarely prove all 10. The ones most likely to **fail** (and therefore the ones to check first) are **1** and **6** -- the closure axioms -- plus **axiom 4** (does a zero vector exist?).

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## ## Subspaces

### ### The Subspace Test (Your Main Proof Tool)

A subset  $H$  of a vector space  $V$  is a **subspace** if and only if three conditions hold:

1. **The zero vector is in  $H$**
2. **Closed under addition:** if  $u, v$  are in  $H$ , then  $u + v$  is in  $H$
3. **Closed under scalar multiplication:** if  $u$  is in  $H$ , then  $cu$  is in  $H$  for any scalar  $c$

This is a shortcut -- if  $H$  lives inside a known vector space  $V$ , you don't need to verify all 10 axioms. The other 7 are inherited automatically from  $V$ .

### ### Why Each Condition Matters

- Condition 1 rules out empty sets and shifted spaces (like a line not through the origin)
- Condition 2 ensures the set doesn't "escape" when you combine vectors
- Condition 3 ensures it doesn't "escape" when you scale

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## ## Proving Something IS a Subspace

**Template for a free response proof:**

```
> Let $H = \{ \dots \}$ be a subset of V .
>
> (1) Zero vector: Show 0 satisfies the defining condition of H .
>
> (2) Closure under addition: Let u, v be arbitrary elements of H . Show $u + v$ also satisfies the condition.
>
```

> **(3) Closure under scalar multiplication:** Let  $u$  be an arbitrary element of  $H$  and  $c$  any scalar. Show  $cu$  satisfies the condition.

>

> Therefore  $H$  is a subspace of  $V$ .

### Worked Example

**Claim:**  $H = \{x \in \mathbb{R}^3 : x_1 + 2x_2 - x_3 = 0\}$  is a subspace of  $\mathbb{R}^3$ .

**(1) Zero vector:** Does  $[0, 0, 0]$  satisfy the equation?  $0 + 0 - 0 = 0$ .  
 $\hat{234}\hat{223}$

**(2) Closure under addition:** Let  $u = [u_1, u_2, u_3]$  and  $v = [v_1, v_2, v_3]$  both be in  $H$ , meaning:

$$- u_1 + 2u_2 - u_3 = 0$$

$$- v_1 + 2v_2 - v_3 = 0$$

Then  $u + v = [u_1+v_1, u_2+v_2, u_3+v_3]$ . Check the condition:

$$(u_1+v_1) + 2(u_2+v_2) - (u_3+v_3) = (u_1 + 2u_2 - u_3) + (v_1 + 2v_2 - v_3) = 0 + 0 = 0$$

$$\hat{234}\hat{223}$$

**(3) Closure under scalar multiplication:** Let  $u$  be in  $H$  and  $c$  any scalar.  $cu = [cu_1, cu_2, cu_3]$ . Check:

$$cu_1 + 2(cu_2) - cu_3 = c(u_1 + 2u_2 - u_3) = c(0) = 0 \hat{234}\hat{223}$$

Therefore  $H$  is a subspace of  $\mathbb{R}^3$ .

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## Proving Something is NOT a Subspace

You only need **one counterexample** to disprove any of the three conditions.

**Most common failure points to check first:**

- Does the zero vector fail the condition? (fastest kill)

- Can you find two vectors in  $H$  whose sum leaves  $H$ ?

- Can you find a vector in  $H$  that leaves  $H$  when scaled by  $-1$ ?

### Worked Example

**Claim:**  $H = \{x \in \mathbb{R}^2 : x_1 + x_2 = 1\}$  is NOT a subspace of  $\mathbb{R}^2$ .

**Disproof (zero vector):** Does  $[0, 0]$  satisfy  $x_1 + x_2 = 1$ ?  $0 + 0 = 0 \hat{211}$   
 1. The zero vector is not in  $H$ , so  $H$  is not a subspace.  $\hat{234}\hat{227}$

That's all you need to write. One failed condition ends the proof.

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## The Big Picture: Span as a Subspace

A critical theorem you should know:

> **The span of any set of vectors is always a subspace.**

This is powerful because it gives you a factory for constructing subspaces. The null space  $\text{Nul}(A)$  and column space  $\text{Col}(A)$  are both subspaces for exactly this reason -- they can each be expressed as spans.

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## Common Exam Traps

| Trap                                                  | Why It Fails                                                       |
|-------------------------------------------------------|--------------------------------------------------------------------|
| $H = \{x \in \mathbb{R}^2 : x_1 x_2 = 0\}$ (the axes) | Closed under scaling, but $[1, 0] + [0, 1] = [1, 1]$ is not in $H$ |

|  $H = \{x \text{ in } \mathbb{R}^2 : x_1 \geq 0\}$  (half-plane) | Fails scalar multiplication: take  $c = -1$  |  
 | Any set defined by an inequality | Almost always fails closure |  
 | Any affine set not through the origin | Always fails the zero vector condition |

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Want me to give you a few proof problems to try on your own, or go deeper on a specific part of this -- like the connection between subspaces and the null/column space of a matrix?

--- You ---

can you go through this example of proving the 3 conditions  $W = \{(x, y, z) \text{ exists in } \mathbb{R}^3 : 2x - y + z = 0\}$

--- Claude ---

Sure, let's walk through it carefully.

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## Claim:  $W = \{(x, y, z) \text{ in } \mathbb{R}^3 : 2x - y + z = 0\}$  is a subspace of  $\mathbb{R}^3$

---

### Condition 1 -- Zero Vector

We need to check that  $(0, 0, 0)$  satisfies the defining equation.

Plug in  $x = 0, y = 0, z = 0$ :

$$2(0) - (0) + (0) = 0$$

The zero vector is in  $W$ .

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### Condition 2 -- Closure Under Addition

Let  $u = (x_1, y_1, z_1)$  and  $v = (x_2, y_2, z_2)$  be arbitrary vectors in  $W$ . That means we know:

$$\begin{aligned} - 2x_1 - y_1 + z_1 &= 0 \\ - 2x_2 - y_2 + z_2 &= 0 \end{aligned}$$

Now form  $u + v = (x_1 + x_2, y_1 + y_2, z_1 + z_2)$  and check whether it satisfies the equation:

$$2(x_1 + x_2) - (y_1 + y_2) + (z_1 + z_2)$$

Rearrange by grouping:

$$= (2x_1 - y_1 + z_1) + (2x_2 - y_2 + z_2)$$

$$= 0 + 0 = 0$$

Since both  $u$  and  $v$  were in  $W$ , their sum satisfies the equation, so  $u + v$  is in  $W$ .

---

### Condition 3 -- Closure Under Scalar Multiplication

Let  $u = (x_1, y_1, z_1)$  be an arbitrary vector in  $W$ , meaning:

$$- 2x_1 - y_1 + z_1 = 0$$

And let  $c$  be any scalar. Form  $cu = (cx_1, cy_1, cz_1)$  and check:

$$2(cx_1) - (cy_1) + (cz_1)$$

Factor out  $c$ :

$$= c(2x_1 - y_1 + z_1)$$

$$= c(0) = 0$$

So  $cu$  is in  $W$  for any scalar  $c$ .

---

### Conclusion

All three conditions of the subspace test hold, so  $W$  is a subspace of  $\mathbb{R}^3$ .

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## What Makes This Example Clean

This works so neatly because  $W$  is defined by a homogeneous linear equation (the right-hand side is 0). That zero is exactly what lets conditions 2 and 3 collapse so cleanly -- you factor and substitute 0.

If the equation were  $2x - y + z = 5$  instead, the zero vector would fail immediately and you'd be done in one line. Any time you see a non-zero constant on the right side, it's not a subspace.

Want to try one on your own and I'll give you feedback, or move on to another concept?

--- You ---

explain this more, and can you render it correctly

--- Claude ---

Sure! Let me break down what's happening in that step algebraically, rendered cleanly.

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Here's a clean step-by-step breakdown of exactly what's happening algebraically in that closure under addition proof, rendered clearly:

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The key insight in steps 3 and 4 is that the algebra is just regrouping by commutativity and associativity -- no tricks. When you expand  $2(x_1 + x_2) - (y_1 + y_2) + (z_1 + z_2)$ , you're allowed to rearrange those terms freely because they're all just real numbers being added and subtracted. That gives you:

$$(2x_1 - y_1 + z_1) + (2x_2 - y_2 + z_2)$$

Which is the magic move, because you've now reorganized the expression into exactly the two facts you were given in step 1 -- both of which equal 0. So the whole thing collapses to 0.

Step 5 is where the assumption from step 1 actually gets used. That's why the

proof starts by letting  $u$  and  $v$  be \*arbitrary\* elements of  $W$  -- it's setting up the two " $= 0$ " substitutions you'll need at the end.

The same exact structure works for closure under scalar multiplication, just with one vector instead of two. Want to walk through that side by side, or try a proof on your own?

--- You ---

since both are equal to 0, we conclude it's equal to 0?

--- Claude ---

Exactly right. Let me make that final logical step crystal clear.

You've shown that:

$$2(x_1 + x_2) - (y_1 + y_2) + (z_1 + z_2) = 2x_1 - y_1 + z_1 + 2x_2 - y_2 + z_2 = 0 + 0 = 0$$

Then you substitute what you know from step 1:

- $2x_1 - y_1 + z_1 = 0$  (because  $u$  is in  $W$ )
- $2x_2 - y_2 + z_2 = 0$  (because  $v$  is in  $W$ )

So the whole expression becomes  $0 + 0 = 0$ .

That final " $= 0$ " is exactly what you needed to prove -- it means  $u + v$  satisfies the equation that defines  $W$ , so  $u + v$  is in  $W$ .

---

The logical flow in one line:

> We need the sum to equal 0. After regrouping, it equals two things that are each already 0 by assumption. Done.

The regrouping step isn't just algebra cleanup -- it's the whole strategy. You're deliberately reshaping the expression until you can see your two assumptions hiding inside it, then replacing them with 0. That pattern shows up in almost every subspace proof involving a linear equation.

Want to try the scalar multiplication condition yourself using the same strategy?

--- You ---

what was my  $u$  and  $v$  again?

--- Claude ---

From earlier in our conversation:

- $u = (x_1, y_1, z_1)$  -- an arbitrary vector in  $W$ , meaning  $2x_1 - y_1 + z_1 = 0$
- $v = (x_2, y_2, z_2)$  -- another arbitrary vector in  $W$ , meaning  $2x_2 - y_2 + z_2 = 0$

The key word being \*arbitrary\* -- we didn't pick specific numbers. We just said "let  $u$  and  $v$  be any two vectors that satisfy the equation defining  $W$ ," which is what makes the proof general and valid for all possible vectors in  $W$ , not just one example.

--- You ---

ohhhh

--- Claude ---

Yeah it clicks once you see it! The "arbitrary but satisfying the condition"

setup is the backbone of basically every subspace proof you'll write.

The pattern is always:

1. **Let**  $u, v$  be arbitrary elements of the set (meaning they satisfy the defining condition)
2. **Use** that condition somewhere in the algebra
3. **Conclude** the result also satisfies the condition

Once you internalize that rhythm, these proofs become very mechanical. Want to try writing the scalar multiplication condition yourself? I'll give you feedback on it.

--- You ---

now how'd I'd prover under scalar multiplication

--- Claude ---

Give it a shot yourself first! Here's your starting point:

Let  $u = (x, y, z)$  be an arbitrary vector in  $W$ , meaning:

$$2x - y + z = 0$$

And let  $c$  be any scalar.

From there, follow the same 5-step rhythm we used for addition -- form  $cu$ , plug it into the equation, regroup, then substitute. Write out what you get and I'll tell you where you're on track or where to adjust.

--- You ---

since we get to the point of  $c(2x - y + z) = 0$  and we know that  $2x - y + z = 0$ , we can assume that any  $c * u$  will be 0

--- Claude ---

Exactly right! You nailed the logic. Let me just tighten the wording slightly for an exam setting:

Rather than "assume," say **conclude** or **therefore** -- because you're not assuming it, you're *proving* it from what you already know:

$$c(2x - y + z) = c(0) = 0$$

You substituted  $2x - y + z = 0$  (which you were given because  $u$  is in  $W$ ), so the whole expression equals 0. That means  $cu$  satisfies the defining equation, so  $cu$  is in  $W$ .

---

The reason this works for **any**  $c$  is exactly your point -- it doesn't matter what  $c$  is, because you're multiplying it by 0. Whether  $c$  is 5, -1000, or 0.003, the result is always 0.

That's actually why homogeneous equations (right-hand side = 0) always define subspaces -- the zero on the right side is what allows that final substitution to work cleanly every time. If the equation were  $2x - y + z = 5$  instead, you'd get  $c(5) = 5c$ , which is only 0 when  $c = 0$ , so it would fail.

You've now proven all three conditions. Want to try writing the full proof from scratch as exam practice?

--- You ---

next example is prove  $S$  is a subspace of  $\mathbb{R}^2$ ,  $S = \{(x, y) \text{ exists in } \mathbb{R}^2 : xy = 0\}$   
It passes the contains 0 vector, but i'm stuck on the addition cause i get  $x_1y_1 + x_1y_2 + x_2y_1 + x_2y_2 = 0$ , i know the first and last products equal 0, but

unsure about the middle. Wait for me for the proof of scala

--- Claude ---

You're thinking exactly right -- that's the key insight for this problem.

$x_1x_2$  and  $x_2x_1$  are the middle cross terms, and you **cannot** guarantee they equal 0. That's actually the point -- this is a case where closure under addition **fails**, so  $S$  is not a subspace.

You can show this with a counterexample. Try:

-  $u = (1, 0)$  -- in  $S$  because  $1^2 = 0$   
 -  $v = (0, 1)$  -- in  $S$  because  $0^2 = 0$   
 -  $u + v = (1, 1)$  -- is this in  $S$ ?  $1^2 = 1 \neq 0$

So two vectors that are individually in  $S$  produce a sum that falls outside  $S$ . Closure under addition fails, therefore  **$S$  is not a subspace**.

---

The deeper reason this fails is that  $xy = 0$  is a **nonlinear** condition -- it's a product of variables, not a linear combination. Subspace proofs work cleanly when the defining equation is linear and homogeneous (like  $2x - y + z = 0$ ). The moment you see multiplication between variables in the definition, suspect it's not a subspace.

Take your time on scalar multiplication and show me what you get!

--- You ---

next topic to cover is demonstrate understanding of linear independence, span, and basis

--- Claude ---

Great, those three are tightly linked -- understanding how they connect to each other is just as important as knowing each definition individually.

Before I dive in, a quick question:

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--- You ---

Q: How comfortable are you with these topics right now?

A: Seen them but shaky

--- Claude ---

Got it. I'll build it up from the definitions with clear examples, same way we did with subspaces.

---

## ## Linear Independence

### ### Definition

A set of vectors  $\{v_1, v_2, \dots, v_n\}$  is **linearly independent** if the only solution to:

$$c_1v_1 + c_2v_2 + \dots + c_nv_n = 0$$

is  $c_1 = c_2 = \dots = c_n = 0$  (called the **trivial solution**).

If any non-zero  $c_i$  exists that still makes it equal 0, the set is **linearly dependent**.

## ### Intuition

Linear dependence means at least one vector in the set is redundant -- it can be written as a combination of the others. Independence means every vector is contributing something new.

## ### How to Test It

Set up  $c_1 v_1 + c_2 v_2 + \dots = 0$ , row reduce, and check if you get only the trivial solution.

## ### Example

Are  $v_1 = (1, 2)$  and  $v_2 = (2, 4)$  linearly independent?

$c_1(1, 2) + c_2(2, 4) = (0, 0)$  gives:

- $c_1 + 2c_2 = 0$
- $2c_1 + 4c_2 = 0$

Row reduce -- the second equation is just 2 times the first, so you get a free variable. Non-trivial solution exists (e.g.  $c_1 = 2, c_2 = -1$ ).  
**\*Linearly dependent\*** --  $v_2$  is just  $2v_1$ .

---

## ## Span

## ### Definition

The **\*span\*** of a set  $\{v_1, \dots, v_n\}$  is the set of **\*all\*** possible linear combinations:

$\text{span}\{v_1, \dots, v_n\} = \{c_1 v_1 + \dots + c_n v_n : c_1, \dots, c_n \text{ are any scalars}\}$

## ### Intuition

Span answers the question: *\*what space can I reach using these vectors?\** Two non-parallel vectors in  $\mathbb{R}^2$  span all of  $\mathbb{R}^2$ . A single vector only spans a line through the origin.

## ### Example

Does  $b = (3, 5)$  lie in  $\text{span}\{(1, 2), (1, 1)\}$ ?

Set up  $c_1(1, 2) + c_2(1, 1) = (3, 5)$ :

- $c_1 + c_2 = 3$
- $2c_1 + c_2 = 5$

Subtract row 1 from row 2:  $c_1 = 2$ , then  $c_2 = 1$ . Solution exists, so **\*yes, b is in the span\***.

---

## ## Basis

## ### Definition

A set of vectors  $B$  is a **\*basis\*** for a vector space  $V$  if:

1.  $B$  is **\*linearly independent\***
2.  $B$  **\*spans\***  $V$

It's the minimal spanning set -- no redundancy, but enough to reach everything.

## ### Key Facts for Exams

- Every basis for a given space has the **\*same number of vectors\*** -- that number is the **\*dimension\***
- The standard basis for  $\mathbb{R}^2$  is  $\{(1, 0), (0, 1)\}$ , for  $\mathbb{R}^3$  is  $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$
- To check if a set is a basis for  $\mathbb{R}^n$ : it must have exactly  $n$  linearly independent vectors

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## ## How They Connect

This is the big picture worth locking in:

> **Span** tells you what space the vectors cover. **Linear independence** tells you there's no waste. **Basis** is when you have both -- full coverage with no redundancy.

Think of it like building a room with LEGOs. Span = can you build the whole room? Independence = are you using any duplicate pieces? Basis = you can build the whole room and every piece is necessary.

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Want to work through an example of checking whether a set of vectors forms a basis, or do you want to try one yourself first?

--- You ---

so lets do the example where we consider the vectors  $\mathbf{v}_1 = 1, 0, 1$ ,  $\mathbf{v}_2 = 0, 1, 1$  and  $\mathbf{v}_3 = 1, 1, 0$

--- Claude ---

Perfect example. The question is: do  $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$  form a basis for  $\mathbb{R}^3$ ?

Remember we need to check **two things** -- linear independence and span. But here's a useful shortcut for  $\mathbb{R}^3$ :

> If you have exactly 3 vectors in  $\mathbb{R}^3$  and they are linearly independent, they automatically span  $\mathbb{R}^3$ . So you only need to check one condition.

So the whole problem reduces to: **are  $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$  linearly independent?**

---

## ## Setting Up

We need to solve  $c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3 = \mathbf{0}$ :

$$c_1(1, 0, 1) + c_2(0, 1, 1) + c_3(1, 1, 0) = (0, 0, 0)$$

This gives the system:

- $c_1 + c_3 = 0$
- $c_2 + c_3 = 0$
- $c_1 + c_2 = 0$

---

## ## Row Reduce

Set up the augmented matrix and row reduce:

$$\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 \end{array}$$

$R_3 - R_1$ :

$$\begin{array}{ccc|c} 1 & 0 & 1 & 0 \end{array}$$

```

[0 1 1 | 0]
[0 1 -1 | 0]
```

```

R3 - R2:

```

```
[1 0 1 | 0]
[0 1 1 | 0]
[0 0 -2 | 0]
```

```

Three pivots, no free variables.

Conclusion

The only solution is $\hat{c}_1 = \hat{c}_2 = \hat{c}_3 = 0$ -- the trivial solution. So $\{\hat{v}_1, \hat{v}_2, \hat{v}_3\}$ are **linearly independent**, and since we have exactly 3 linearly independent vectors in $\mathbb{R}^3$, they **span $\mathbb{R}^3$** .

Therefore $\{\hat{v}_1, \hat{v}_2, \hat{v}_3\}$ is a **basis for $\mathbb{R}^3$** .

Now try this one yourself: what if $\hat{v}_3 = (1, 1, 2)$ instead? Set up the row reduction and tell me what you get.

--- You ---

does it span $\mathbb{R}^3$?

--- Claude ---

Yes -- and that conclusion follows directly from the linear independence we just proved.

Here's the logic chain:

1. $\mathbb{R}^3$ has dimension 3, meaning any basis needs exactly 3 vectors
2. We have exactly 3 vectors
3. They are linearly independent

Those three facts together guarantee they span $\mathbb{R}^3$. You don't need to do any extra work.

The intuition is that if you have 3 linearly independent vectors in $\mathbb{R}^3$, there's no "direction" left uncovered -- each vector is pointing somewhere genuinely new that the others can't reach, and 3 independent directions is exactly what you need to fill all of $\mathbb{R}^3$.

A useful thing to remember for exams:

Situation	Conclusion
n linearly independent vectors in $\mathbb{R}^n$	Automatically spans $\mathbb{R}^n$
Fewer than n vectors	Cannot span $\mathbb{R}^n$
More than n vectors	Cannot be linearly independent

So for square cases like this one, independence and span are two sides of the same coin -- prove one and you get the other for free.

Want to try the $\hat{v}_3 = (1, 1, 2)$ version I gave you, or move on to a different aspect of basis?

--- You ---

conclude whether it is a basis and explain why, then express standard basis vector $e_1 = (1, 0, 0)$

--- Claude ---

Give it a shot first! You already have the tools from the original example.

Set up $c_1 v_1 + c_2 v_2 + c_3 v_3 = 0$ with the new $v_3 = (1, 1, 2)$ and row reduce. Tell me what you get and whether you hit any free variables -- then we'll conclude together and move straight into expressing e_1 .

--- You ---

why 1,1,2?

--- Claude ---

That was just a practice problem I made up to give you something to try on your own -- the idea being that changing v_3 from $(1, 1, 0)$ to $(1, 1, 2)$ might change whether the set is still a basis.

But we don't have to do that one. Want to just move straight into expressing $e_1 = (1, 0, 0)$ as a linear combination of the original v_1, v_2, v_3 from our example?

--- You ---

yes please

--- Claude ---

Alright. The question is: can we find scalars c_1, c_2, c_3 such that:

$$c_1 v_1 + c_2 v_2 + c_3 v_3 = e_1$$

Written out:

$$c_1 (1, 0, 1) + c_2 (0, 1, 1) + c_3 (1, 1, 0) = (1, 0, 0)$$

This gives the system:

- $c_1 + c_3 = 1$
- $c_2 + c_3 = 0$
- $c_1 + c_2 = 0$

Row Reduce

$$\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 \end{array}$$

R3 - R1:

$$\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & -1 & -1 \end{array}$$

R3 - R2:

$$\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & -2 & -1 \end{array}$$

...

From row 3: $c_3 = 1/2$

Back substitute into row 2: $c_2 + 1/2 = 0 \Rightarrow c_2 = -1/2$

Back substitute into row 1: $c_1 + 1/2 = 1 \Rightarrow c_1 = 1/2$

Conclusion

$e_1 = \frac{1}{2}v_1 - \frac{1}{2}v_2 + \frac{1}{2}v_3$

Check: $\frac{1}{2}(1,0,1) - \frac{1}{2}(0,1,1) + \frac{1}{2}(1,1,0) = (1/2, 0, 1/2) + (0, -1/2, -1/2) + (1/2, 1/2, 0) = (1, 0, 0)$

This is exactly what it means for $\{v_1, v_2, v_3\}$ to span $\mathbb{R}^3$ -- any vector in $\mathbb{R}^3$, including the standard basis vectors, can be expressed as a linear combination of them. Want to try expressing $e_2 = (0,1,0)$ yourself?

--- You ---

so if the vectors are linearly independent, that means they are a basis, and have a span?

--- Claude ---

Almost -- just needs one small refinement. Linear independence alone doesn't make something a basis. You also need to know **what space** you're working in.

The full picture:

Vectors	Linearly Independent?	Spans $\mathbb{R}^3$?	Basis for $\mathbb{R}^3$?
3 vectors in $\mathbb{R}^3$	Yes	Yes	Yes
3 vectors in $\mathbb{R}^3$	No	Maybe	No
2 vectors in $\mathbb{R}^3$	Yes	No	No
4 vectors in $\mathbb{R}^3$	Can't be	Yes	No

The shortcut that applies to **our specific example** is:

> If you have **exactly n** linearly independent vectors in $\mathbb{R}^n$, then yes -- they automatically form a basis and span $\mathbb{R}^n$.

The count has to match the dimension. Two linearly independent vectors in $\mathbb{R}^3$ are independent but can only span a **plane**, not all of $\mathbb{R}^3$ -- so they wouldn't be a basis for $\mathbb{R}^3$.

So the complete statement is:

> Linear independence + correct count $\Rightarrow$ basis $\Rightarrow$ spans the space

Does that distinction make sense? Want to move on to another concept or keep drilling this one?

--- You ---

this makes sense, so it has to have 3 linearly independent vectors and must span $\mathbb{R}^3$ in order to be a basis for $\mathbb{R}^3$?

--- Claude ---

Exactly -- but remember those two conditions are actually linked when the count matches. So in practice for exams:

> If you have exactly 3 vectors in $\mathbb{R}^3$, **you only need to check linear independence**. If they're independent, span is guaranteed for free. If they're dependent, it fails both.

You'd only need to explicitly verify span if someone handed you more than 3 vectors and asked if a subset forms a basis, or if the space isn't $\mathbb{R}^3$.

So the exam workflow for "is this a basis for $\mathbb{R}^3$?" is:

1. Count the vectors -- if it's not 3, immediately no
2. Row reduce and check for free variables -- if none, linearly independent
â\206\222 **it's a basis**

That's it. Two steps and you're done.

Ready to move on to the next topic, or do you want one more example to solidify this?